

COMPSCI 5100

ML & AI for Data Science

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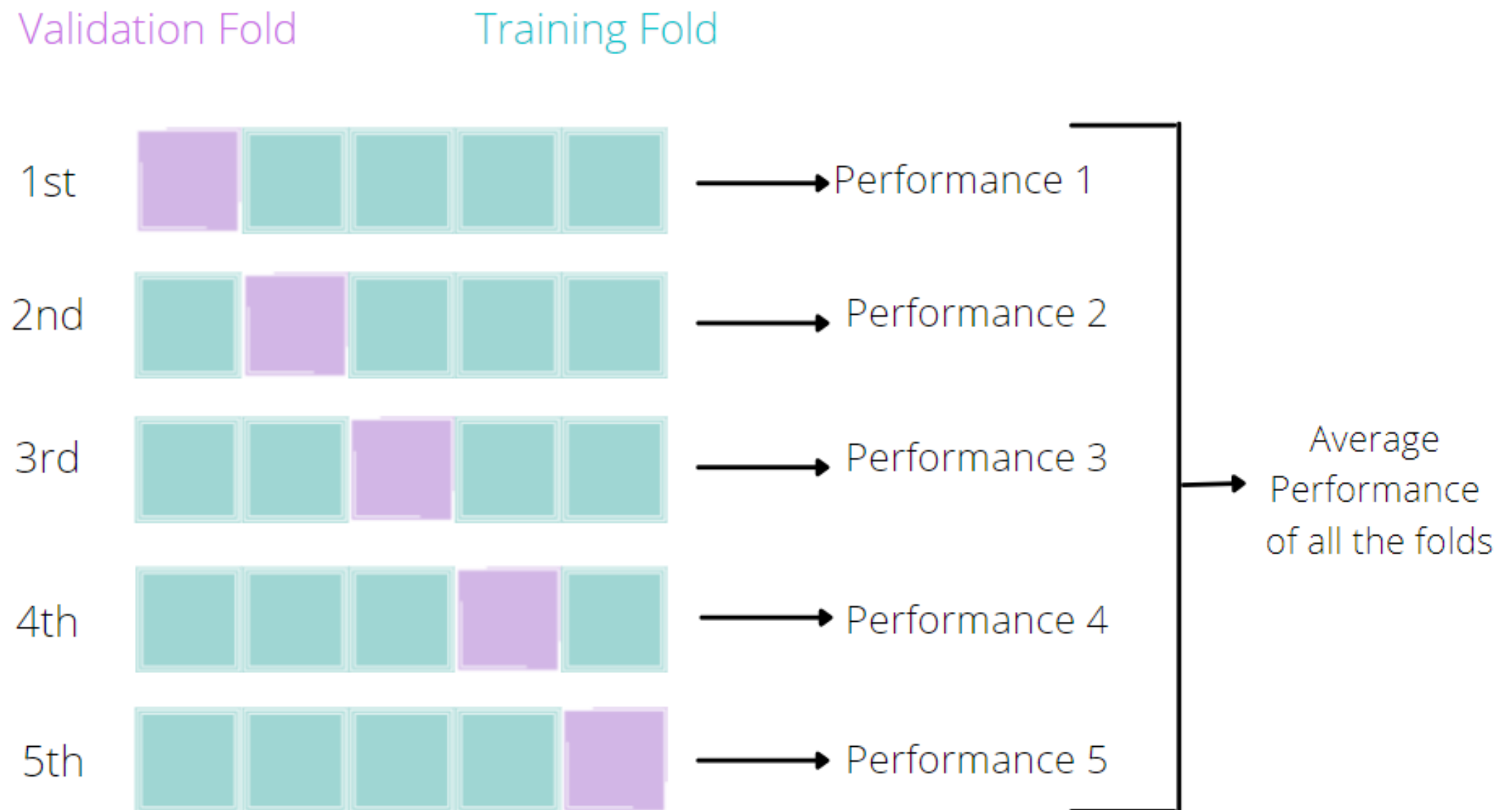
Classification: Part III

Performance evaluation

- **Evaluation strategy**
 - How to split data for training and testing
- **Evaluation metrics**
 - How to measure performance accurately
- **Benchmarking**
 - Compare results against other 'known' results

Evaluation strategy

- **Cross validation**



Evaluation strategy

- **Leave one subject out**
 - Particularly useful for classification tasks involving human-centric data

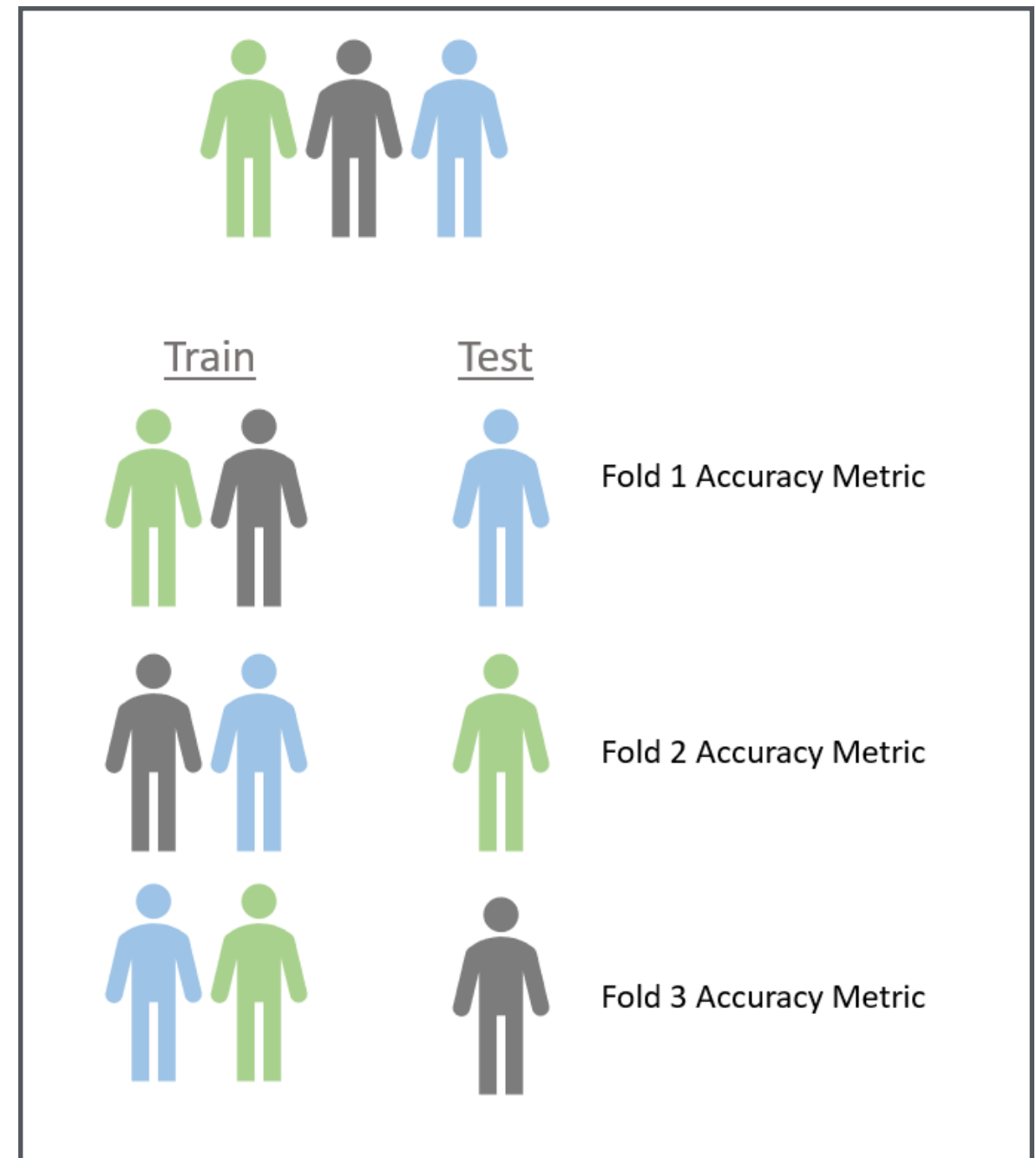
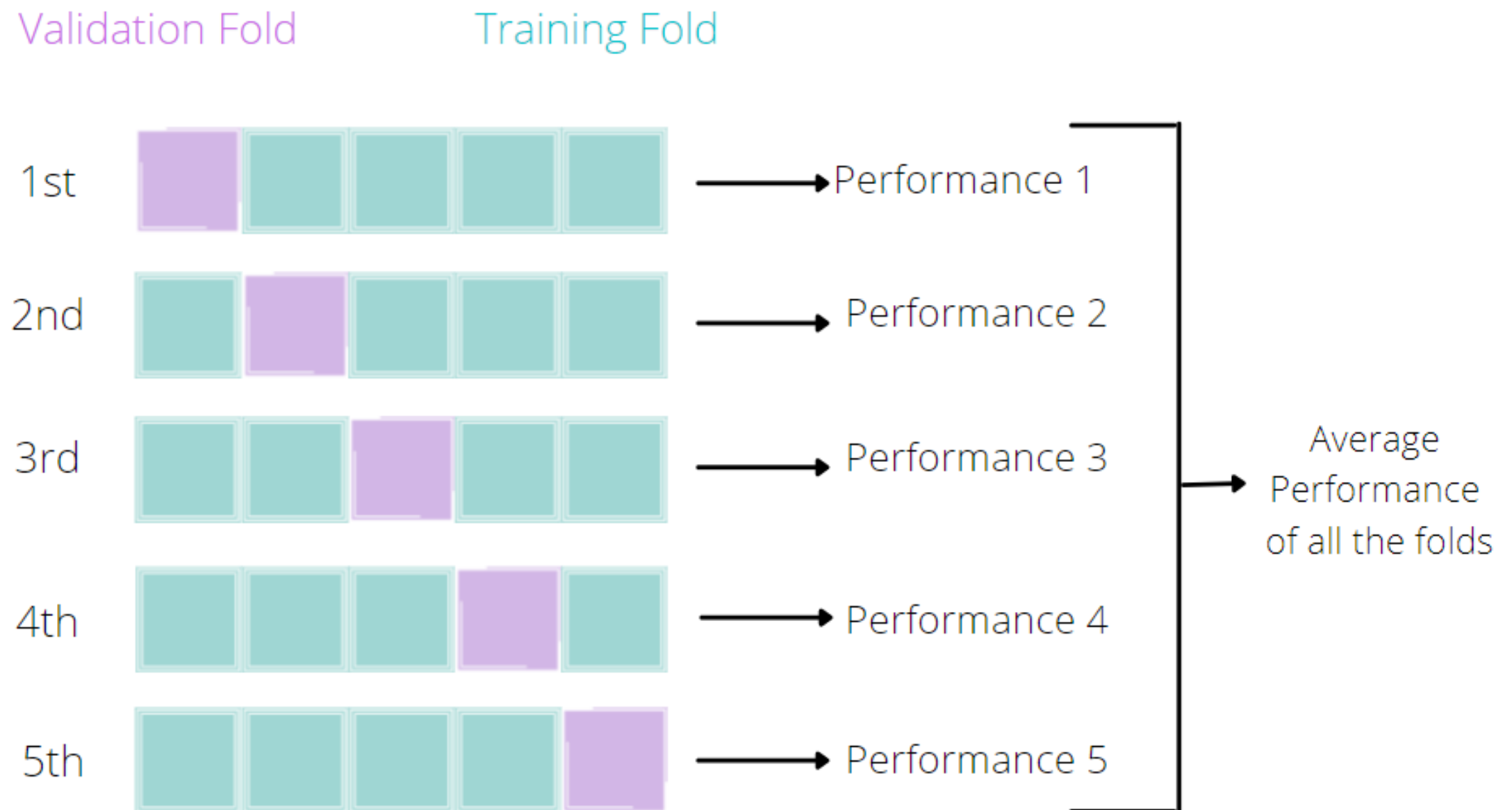


image from: medium.com

Evaluation strategy

- **Cross validation**



Evaluation strategy

- **Random train:test splits**
 - Particularly useful for very large datasets
 - CV may be difficult
 - Randomly choose 70-80% data for training and rest for testing



Performance metrics

- Performance **metrics** are important to
 - Compare performances of multiple classifiers
 - Compare performance of the same classifier under different conditions
 - Tune hyperparameters
- No metric is perfect; each gives you some insights
- Practical tip: **Use multiple evaluation metrics**

Accuracy

- **Accuracy** = $\frac{\text{Number of correctly classified samples}}{\text{Total number of test samples}}$
- Often expressed in %
- Simple, intuitive, widely used

Disadvantage: Doesn't take into account class imbalance:

- ▶ We're building a classifier to detect a rare disease.
- ▶ Assume only 1% of population is diseased.
- ▶ Diseased: $t = 1$
- ▶ Healthy: $t = 0$
- ▶ What if we always predict healthy? ($t = 0$)
- ▶ Accuracy 99%
- ▶ But classifier is rubbish!

[Content from Dr. Ke Yuan's slide]

Weighted accuracy (WA):
Accuracies computed per class,
averaged across all classes

Confusion matrix

		True	
		1	0
Predicted	1	True Positives (TP)	False Positives (FP)
	0	False Negatives (FN)	True Negatives (TN)

		True class											
		...	10	11	12	13	14	15	16	18	18	19	20
Predicted class	1	...	4	2	0	2	10	4	7	1	12	7	47
	2	...	0	0	4	18	7	8	2	0	1	1	3
	3	...	0	0	1	0	1	0	1	0	0	0	0
	4	...	1	0	1	28	3	0	0	0	0	0	0
	...												
	16	...	3	2	2	5	17	4	376	3	7	2	68
	17	...	1	0	9	0	3	1	3	325	3	95	19
	18	...	2	1	0	2	6	2	1	2	325	4	5
	19	...	8	4	8	0	10	21	1	16	19	185	7
	20	...	0	0	1	0	1	1	2	4	0	1	92

- ▶ Algorithm is getting 'confused' between classes 20 and 16, and 19 and 17.
 - ▶ 17: talk.politics.guns
 - ▶ 19: talk.politics.misc
 - ▶ 16: talk.religion.misc
 - ▶ 20: soc.religion.christian
- ▶ Maybe these should be just one class?
- ▶ Maybe we need more data in these classes?
- ▶ Confusion matrix helps us direct our efforts to improving the classifier.

[Content from Dr. Ke Yuan's slide]

Precision

		True	
		1	0
Predicted	1	True Positives (TP)	False Positives (FP)
	0	False Negatives (FN)	True Negatives (TN)

Example:

1: Diseased 0: Healthy

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

- Among all people classified as 'diseased', how many are actually diseased
- Perfect precision = no FP
- Higher the better

Recall or Sensitivity

		True	
		1	0
Predicted	1	True Positives (TP)	False Positives (FP)
	0	False Negatives (FN)	True Negatives (TN)

Example:

1: Diseased 0: Healthy

$$\text{Sensitivity } S_e = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

- Among all diseased people, how many are correctly identified
- Sensitivity = recall
- Perfect recall = no FN
- Higher the better

Specificity

		True	
		1	0
Predicted	1	True Positives (TP)	False Positives (FP)
	0	False Negatives (FN)	True Negatives (TN)

Example:

1: Diseased 0: Healthy

$$\text{Specificity } S_p = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

- Among all healthy people, how many are classified as healthy.
- Higher the better

Optimizing sensitivity and specificity

- ▶ We would like both to be as high as possible.
- ▶ Often increasing one will decrease the other.
- ▶ Balance will depend on application:
- ▶ e.g. diagnosis:
 - ▶ We can probably tolerate a decrease in specificity (healthy people diagnosed as diseased)....
 - ▶ ...if it gives us an increase in sensitivity (getting diseased people right).

[Slide courtesy: Dr. Ke Yuan]

ROC

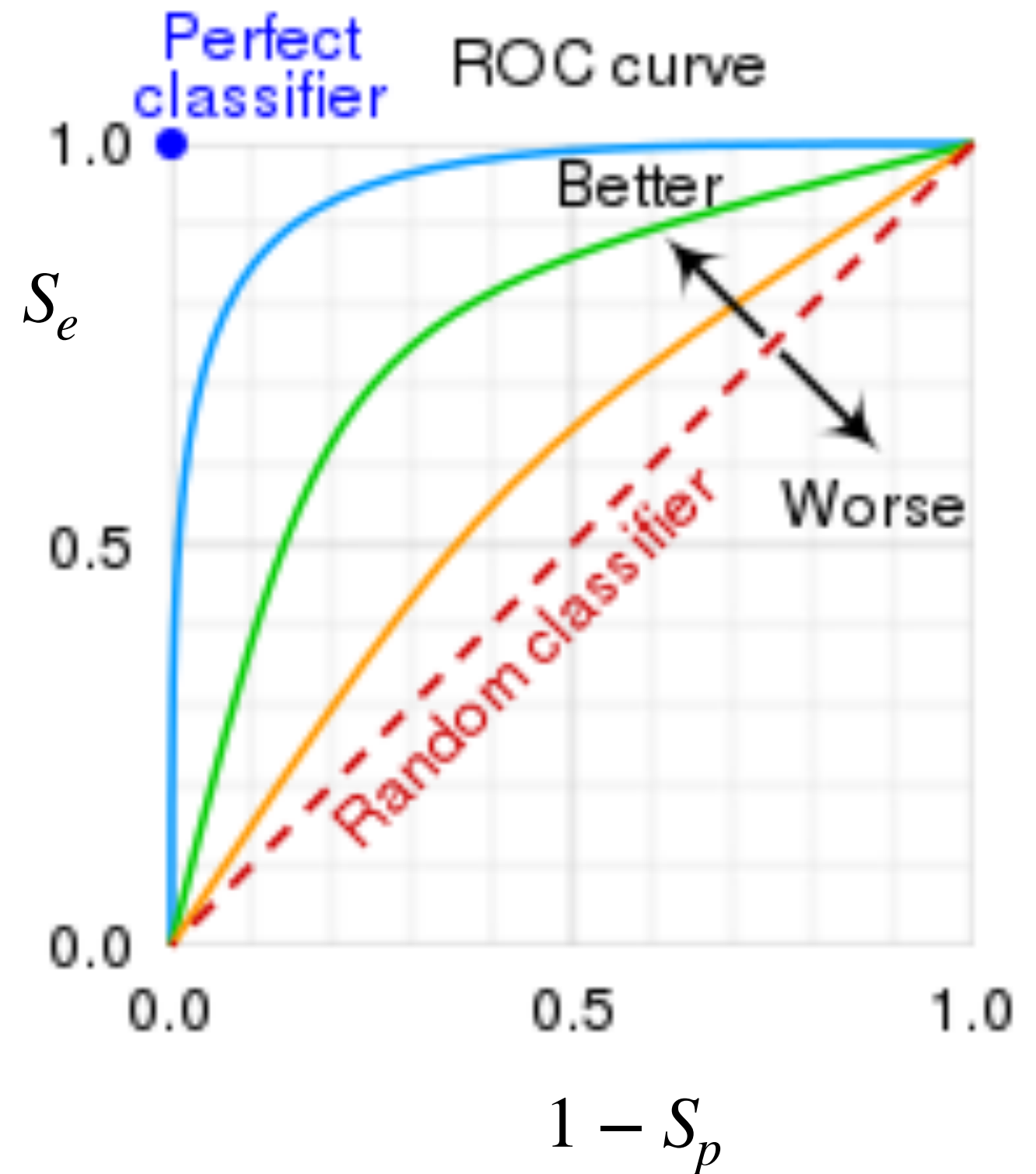
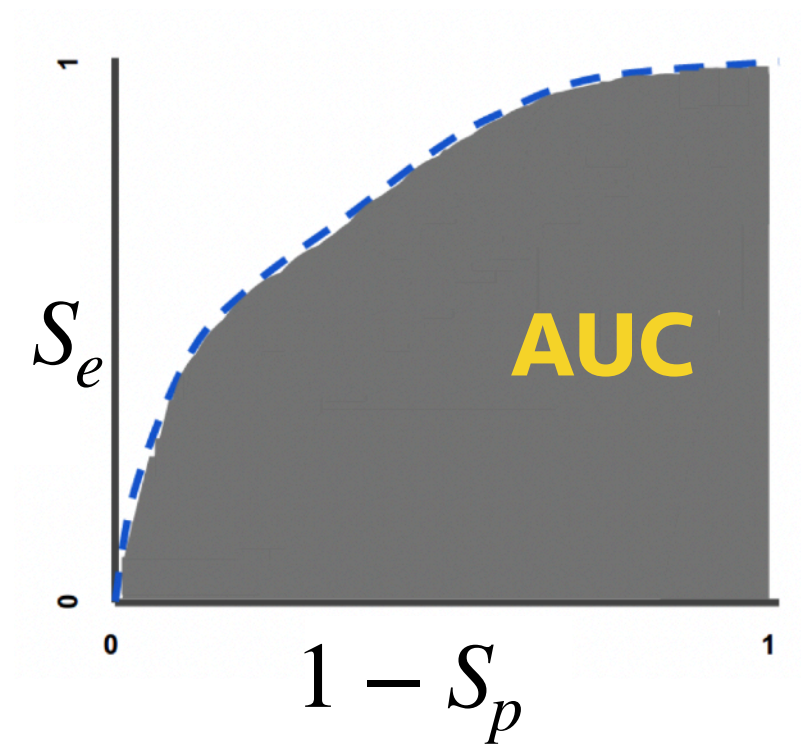
- ▶ Many classification algorithms involve setting a threshold.
- ▶ e.g. Logistic Regression:

$$p(t_{new} = 1 | \mathbf{x}_{new}, \mathbf{w}) > 0.5$$

- ▶ Implies a threshold of zero (sign function)
- ▶ However, we could use any threshold we like....
- ▶ The *Receiver Operating Characteristic (ROC) curve* shows how S_e and $1 - S_p$ vary as the threshold changes.

[Slide courtesy: Dr. Ke Yuan]

ROC, AUC



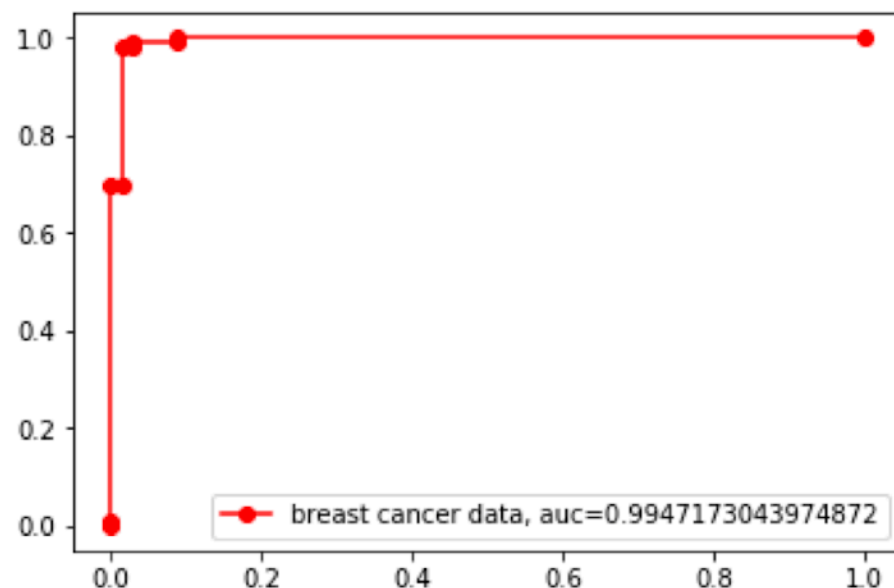
[Image from Wikipedia]


```
from sklearn.model_selection import train_test_split
from sklearn import metrics
from sklearn.datasets import load_breast_cancer

breast_cancer = load_breast_cancer()
X = breast_cancer.data
t = breast_cancer.target

X_train, X_test, y_train, y_test = train_test_split(X,t,test_size=0.30, random_state=123)
clf1 = LogisticRegression().fit(X_train, y_train)

y_pred1 = clf1.predict(X_test)
y_pred_proba1 = clf1.predict_proba(X_test)[:,-1]
fpr1, tpr1, _ = metrics.roc_curve(y_test, y_pred_proba1)
auc1 = metrics.roc_auc_score(y_test, y_pred_proba1)
plt.plot(fpr1,tpr1,'ro-',label="breast cancer data, auc="+str(auc1))
plt.legend(loc=4)
plt.show()
```



Try it on a breast cancer dataset
Plot ROC of a Logistic Regression model

F1

- Metric combining Precision and Recall

- $$\mathbf{F1} = \frac{2}{\text{Precision}^{-1} + \text{Recall}^{-1}} = \frac{2TP}{2TP + FP + FN}$$

- Bounded between 0 to 1
- Higher the better